5 Problems

1.* Suppose n is of the form $n = 4^a(8k + 7)$, where a and k are positive integers. Show that n cannot be written as the sum of three-squares. The converse, that every n that is *not* of that form *can* be written as the sum of three-squares, is a difficult theorem of Legendre and Gauss.

2. Let $\mathrm{SL}_2(\mathbb{Z})$ denote the set of 2×2 matrices with integer entries and determinant 1, that is,

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \text{ and } ad - bc = 1 \right\}.$$

This group acts on the upper half-plane by the fractional linear transformation $g(\tau) = (a\tau + b)/(c\tau + d)$. Together with this action comes the so-called fundamental domain $\mathcal{F}_1$ in the complex plane defined by

$$\mathcal{F}_1 = \{\tau \in \mathbb{C} : |\tau| \geq 1, \quad |\mathrm{Re}(\tau)| \leq 1/2 \text{ and } |\mathrm{Im}(\tau)| \geq 0\}.$$

It is illustrated in Figure 3.

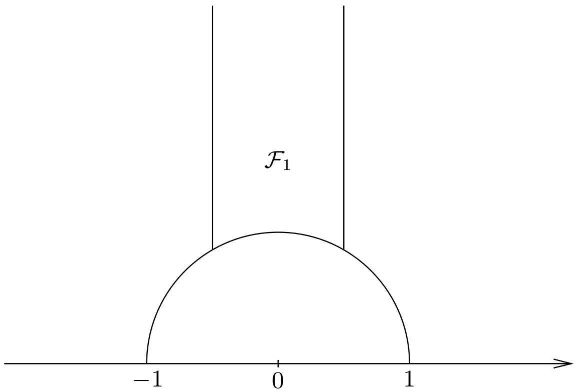

Figure 3. The fundamental domain $\mathcal{F}_1$

Consider the two elements in $\mathrm{SL}_2(\mathbb{Z})$ defined by $S(\tau) = -1/\tau$ and $T_1(\tau) = \tau + 1$. These correspond (for example) to the matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

respectively. Let $\mathfrak{g}$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by S and T_1.

(a) Show that for every $\tau \in \mathbb{H}$ there exists $g \in \mathfrak{g}$ such that $g(\tau) \in \mathcal{F}_1$.

(b) We say that two points τ and τ' are congruent if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g(\tau) = w$. Prove that if $\tau, w \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re}(\tau) =$

$\pm 1/2$ and $\tau' = \tau \mp 1$ or $|\tau| = 1$ and $\tau' = -1/\tau$. [Hint: Say $\tau' = g(\tau)$. Why can one assume that $\mathrm{Im}(\tau') \geq \mathrm{Im}(\tau)$, and therefore $|c\tau + d| \leq 1$? Now consider separately the possibilities $c = -1$, $c = 0$, or $c = 1$.]

(c) Prove that S and T_1 generate the modular group in the sense that every fractional linear transformation corresponding to $g \in \mathrm{SL}_2(\mathbb{Z})$ is a composition of finitely many S's and T_1's, and their inverses. Strictly speaking, the matrices associated to S and T_1 generate the projective special linear group $\mathrm{PSL}_2(\mathbb{Z})$, which equals $\mathrm{SL}_2(\mathbb{Z})$ modulo $\pm I$. [Hint: Observe that $2i$ is in the interior of $\mathcal{F}_1$. Now map $g(2i)$ back into $\mathcal{F}_1$ by using part (a). Use part (b) to conclude.]

3. In this problem, consider the group G of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integer entries, determinant 1, and such that a and d have the same parity, b and c have the same parity, and c and d have opposite parity. This group also acts on the upper half-plane by fractional linear transformations. To the group G corresponds the fundamental domain $\mathcal{F}$ defined by $|\tau| \geq 1$, $|\mathrm{Re}(\tau)| \leq 1$, and $\mathrm{Im}(\tau) \geq 0$ (see Figure 1). Also, let

$$S(\tau) = -1/\tau \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T_2(\tau) = \tau + 2 \leftrightarrow \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Prove that every fractional linear transformation corresponding to $g \in G$ is a composition of finitely many S, T_2 and their inverses, in analogy with the previous problem.

4. Let G denote the group of matrices given in the previous problem. Here we give an alternate proof of Theorem 3.4, that states that a function in $\mathbb{H}$ which is holomorphic, bounded, and invariant under G must be constant.

(a) Suppose that $f : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and that there exists a sequence of complex numbers $\tau_k = x_k + iy_k$ such that

$$f(\tau_k) = 0, \qquad \sum_{k=1}^{\infty} y_k = \infty, \qquad 0 < y_k \leq 1, \qquad \text{and} \qquad |x_k| \leq 1.$$

Then $f = 0$. [Hint: When $x_k = 0$ see Problem 5 in Chapter 8.]

(b) Given two relatively prime integers c and d with different parity, show that there exist integers a and b such that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$. [Hint: All the solutions of $xc + dy = 1$ take the form $x_0 + dt$ and $y_0 - ct$ where x_0, y_0 is a particular solution and $t \in \mathbb{Z}$.]

(c) Prove that $\sum 1/(c^2 + d^2) = \infty$ where the sum is taken over all c and d that are relatively prime and of opposite parity. [Hint: Suppose not, and prove that $\sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ where the sum is over all relatively prime integers a and b. To do so, note that if a and b are both odd and relatively prime, then the two numbers c and d defined by $c = (a + b)/2$

and $d = (a-b)/2$ are relatively prime and of opposite parity. Moreover, $c^2 + d^2 \leq A(a^2+b^2)$ for some universal constant A. Therefore

$$\sum_{n\neq 0} \frac{1}{n^2} \sum_{(a,b)=1} \frac{1}{a^2+b^2} < \infty,$$

hence $\sum 1/(k^2+\ell^2) < \infty$, where the sum is over all integers k and ℓ such that $k, \ell \neq 0$. Why is this a contradiction?]

(d) Prove that if $F : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and invariant under G, then F is constant. [Hint: Replace $F(\tau)$ by $F(\tau) - F(i)$ so that we can assume $F(i) = 0$ and prove $F = 0$. For each relatively prime c and d with opposite parity, choose $g \in G$ so that $g(i) = x_{c,d} + i/(c^2+d^2)$ with $|x_{c,d}| \leq 1$.]

5.* In Chapter 9 we proved that the Weierstrass $\wp$ function satisfies the cubic equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2 = 60E_4$, $g_3 = 140E_6$, with E_k is the Eisenstein series of order k. The discriminant of the cubic $y^2 = 4x^3 - g_2x - g_3$ is defined by $\triangle = g_2^3 - 27g_3^2$. Prove that

$$\triangle(\tau) = (2\pi)^{12}\eta^{24}(\tau) \qquad \text{for all } \tau \in \mathbb{H}.$$

[Hint: $\triangle$ and η^{24} satisfy the same transformation laws under $\tau \mapsto \tau+1$ and $\tau \mapsto -1/\tau$. Because of the fundamental domain described in Problem 2, it suffices then to investigate the behavior at the only cusp, which is at infinity.]

6.* Here we will deduce the formula for $r_8(n)$, which counts the number of representations of n as a sum of eight squares. The method is parallel to that of $r_4(n)$, but the details are less delicate.

Theorem: $r_8(n) = 16\sigma_3^*(n)$.

Here $\sigma_3^*(n) = \sigma_3(n) = \sum_{d|n} d^3$, when n is odd. Also, when n is even

$$\sigma_3^*(n) = \sum_{d|n} (-1)^d d^3 = \sigma_3^e(n) - \sigma_3^o(n),$$

where $\sigma_3^e(n) = \sum_{d|n,\ d \text{ even}} d^3$ and $\sigma_3^o(n) = \sum_{d|n,\ d \text{ odd}} d^3$.

Consider the appropriate Eisenstein series

$$E_4^*(\tau) = \sum \frac{1}{(n+m\tau)^4},$$

where the sum is over integers n and m with opposite parity. Recall the standard Eisenstein series

$$E_4(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^4}.$$

Notice that the series defining E_4^* is absolutely convergent, in distinction to $E_2^*(\tau)$, which arose when considering $r_4(n)$. This makes some of the considerations below quite a bit simpler.

(a) Prove that $r_8(n) = 16\sigma_3^*(n)$ is equivalent to the identity $\theta(\tau)^8 = 48\pi^{-4}E_4^*(\tau)$. [Hint: Use the fact that $E_4(\tau) = 2\zeta(4) + \frac{(2\pi)^4}{3}\sum_{k=1}^{\infty}\sigma_3(k)e^{2\pi ik\tau}$ and $\zeta(4) = \pi^4/90$.]

(b) Note that $E_4^*(\tau) = E_4(\tau) - 2^{-4}E_4((\tau-1)/2)$.

(c) $E_4^*(\tau+2) = E_4^*(\tau)$.

(d) $E_4^*(\tau) = \tau^{-4}E_4^*(-1/\tau)$.

(e) $(48/\pi^4)E_4^*(\tau) \to 1$ as $\tau \to \infty$.

(f) $|E_4^*(1-1/\tau)| \approx |\tau|^4|e^{2\pi i\tau}|$, as $\mathrm{Im}(\tau) \to \infty$. [Hint: Verify that $E_4^*(1-1/\tau) = \tau^4(E_4(\tau) - E_4(2\tau))$.]

Since $\theta(\tau)^8$ satisfies properties similar to (c), (d), (e) and (f) above, it follows that the invariant function $48\pi^{-4}E_4^*(\tau)/\theta(\tau)^8$ is bounded and hence a constant, which must be 1. This gives the desired result.

Appendix A: Asymptotics

On the numerical computation of the definite integral $\int_w \cos \frac{\pi}{2}(w^3 - m.w)$, between the limits 0 and $\frac{1}{0}$.

The simplicity of the form of this differential coefficient induces me to suppose that the integral may possibly be expressible by some of the integrals whose values have been tabulated. After many attempts however, I have not succeeded in reducing it to any known integral: and I have therefore computed its value by actual summation to a considerable extent and by series for the remainder.

G. B. Airy, 1838

In a number of problems in analysis the solution is given by a function whose explicit calculation is not tractable. Often a useful substitute (and the only recourse) is to study the asymptotic behavior of this function near the point of interest. Here we shall investigate several related types of asymptotics, where the ideas of complex analysis are of crucial help. These typically center about the behavior for large values of the variable s of an integral of the form

$$I(s) = \int_a^b e^{-s\Phi(x)}\, dx. \tag{1}$$

We organize our presentation by formulating three guiding principles.

(i) **Deformation of contour**. The function Φ is in general complex-valued, therefore, for large s the integrand in (1) may oscillate rapidly, so that the resulting cancellations mask the true behavior of $I(s)$. When Φ is holomorphic (which is often the case) one can hope to change the contour of integration so that as far as possible, on the new contour Φ is essentially real-valued. If this is possible, one can then hope to read off the behavior of $I(s)$ in a rather direct manner. This idea will be illustrated first in the context of Bessel functions.

(ii) **Laplace's method**. In the case when Φ is real-valued on the contour and s is positive, the maximum contribution to $I(s)$ comes